Introduction to Statistical Methods(ISM) (AIMLC ZC418)

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Session 2
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➢**Module 1** : Basic Probability & Statistics

Quick Recap

➢ Three commonly used measures of central tendency:

1. Mean
2. Median
3. Mode

Day	1	2	3	4	5	6	7	8	9	10
Time (min)	39	29	43	52	39	44	40	31	44	35

$$\overline{X} = \frac{\text{Sum of the values}}{\text{Number of values}}$$

$$\overline{X} = \frac{\sum_{i=1}^{n} X_i}{n}$$

$$\overline{X} = \frac{39 + 29 + 43 + 52 + 39 + 44 + 40 + 31 + 44 + 35}{10}$$

$$\overline{X} = \frac{396}{10} = 39.6$$

Example:

➢ A systems manager in charge of a company s network keeps track of the number of server failures that occur in a day. Compute the mode for the following data, which represents the number of server failures in a day for the past two weeks:.

1 3 0 3 26 2 7 4 2 3 3 6 3

➢ Because 3 appears five times, more times than any other value, the mode is 3. Thus, the systems manager can say that the most common occurrence is having three server failures in a day.

Quick Recap

- **Example-1**: The three-year annualized returns for the seven small-cap growth funds with low risk are ranked from the smallest to the largest:

  The Median is the 4th position value i.e. 22.4.

- **Example-2**: Consider the time taken for reaching office from home in minutes arranged from lowest to highest

29	31	35	39	39	40	43	44	44	52

- The number of observations are 10. Hence the median is the average of values at the 5th and 6th Position, i.e. average of 39 and 40 = **39.5**

Quick Recap

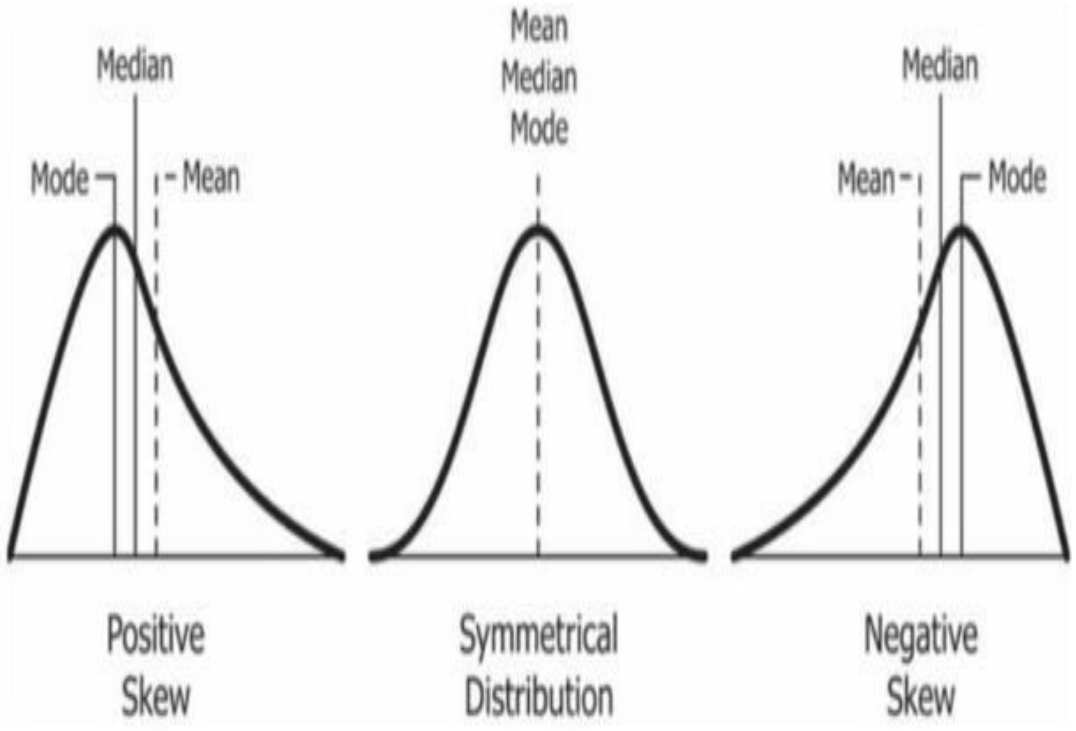
- Skewed

  - Negatively (Left):mean < median<Mode
    - Distributions with a left skew have long left tails;
  - Positively (Right) : mean > median>Mode
    - Distributions with a right skew have long right tails.

- Symmetry: Mean=Median=Mode

- Bimodal : has two distinct modes

- Multi-modal : has more than 2 distinct modes



Note: In science, an empirical relationship is a relationship that is supported by experiment and observation but not necessarily supported by theory. For moderately skewed/unskew data, an empirical relationship is observed as Mode = 3Median-2Mean. Further Mean is applied for unskew/moderately skewed data whereas median is a preferred measure for heavily skewed data.

Measures of variability/dispersion/spread

- Measures of Variability:
  - The Range
  - The Variance
  - The Standard Deviations
  - Quartile deviation

Example: For a set of scores: 7, 2, 7, 6, 5, 6, 2
Range = Highest Score minus Lowest score = 7 - 2 = 5

Variance equals mean (average) squared deviation (distance) of the scores from the mean

$$\text{Variance} = \frac{\text{sum of squared deviations}}{\text{number of scores}}$$

$$SS = \sum (X - \mu)^2$$

Sl. No.	X_1
1	2
2	8
3	5
4	3
5	7
6	8
7	5
8	2
9	5
Total	45

$$\overline{X} = \frac{\sum_{1}^{9} x_i}{N} = \frac{45}{9} = 5$$

$$S = \sqrt{\frac{\sum (X - \overline{X})^2}{n-1}}$$

$$\mathbf{S = \sqrt{\frac{44}{8}} = 2.345}$$

Five point summary of Data

The five number summary of data includes 5 items:

- **Minimum**.
- **Q1** (the first quartile, or the 25% mark).
- **Median**.
- **Q3** (the third quartile, or the 75% mark).
- **Maximum**.

Interquartile range (IQR) & Quartile Deviation (QD)

- It is measure of Variation
- Also Known as Mid-spread : Spread in the Middle 50%
- Difference Between Third & First Quartiles:
- Not Affected by Extreme Values
- Interquartile Range = IQR = $Q_3 - Q_1$

Data in Ordered Array: 11 12 13 16 16 17 17 18 21

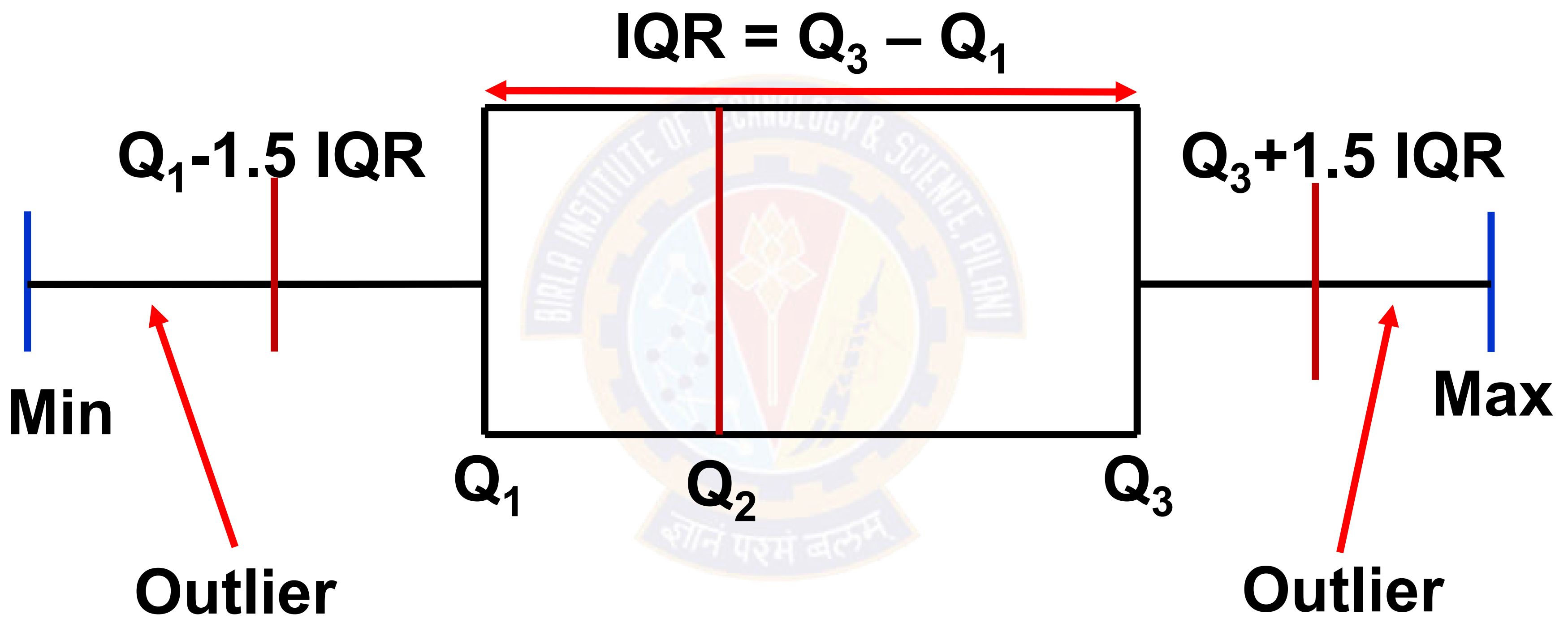
$$\text{Position of } Q_1 = \frac{1 \bullet (9+1)}{4} = 2.50, \qquad Q_1 = 12.5$$

$$\text{Position of } Q_3 = \frac{3 \bullet (9+1)}{4} = 7.50, \qquad Q_3 = 17.5$$

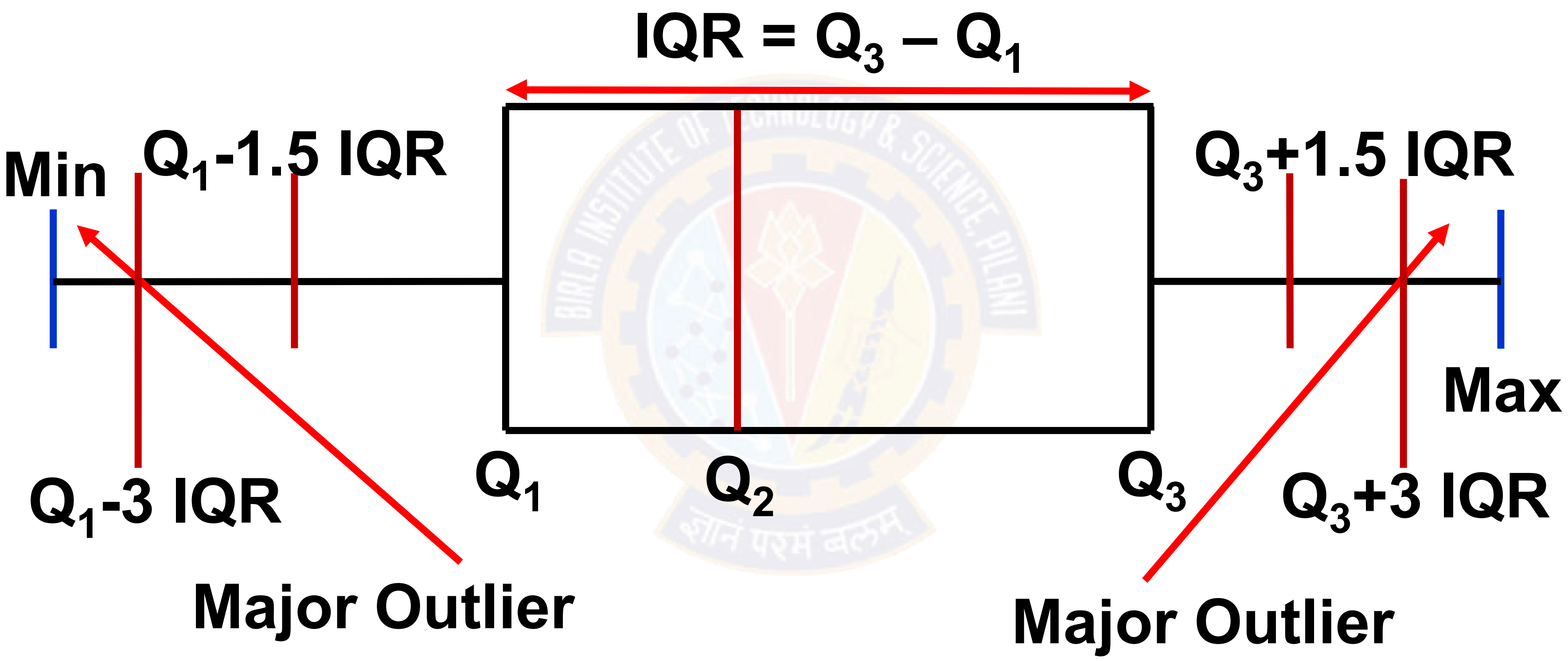
Interquartile Range = IQR= $Q_3 - Q_1$ = 17.5 - 12.5 = 5

QD = IQR/2 = 5/2 = 2.5 (variation among the data in the central 50% of a data set)

Box and Whisker plot



Box and Whisker plot



Detection of outliers

- Data in Ordered Array of the previous data: 11 12 13 16 16 17 17 18
- Q_1-1.5 IQR =5, 11 12 13 16 16 17 17 18, Q_3+1.5 IQR = 25
- As all the above data points lie in between Q_1-1.5 IQR and Q_3+1.5 IQR , the above data is free from Outliers/Extreme Values

Potential outliers

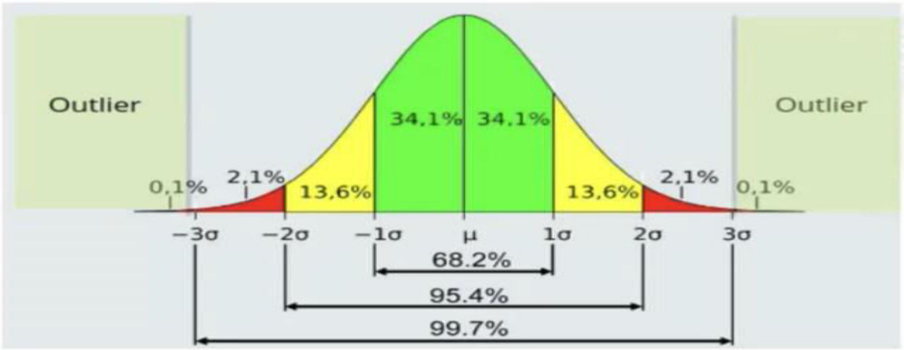
❖ The lower limit and upper limit of a data set are given by:

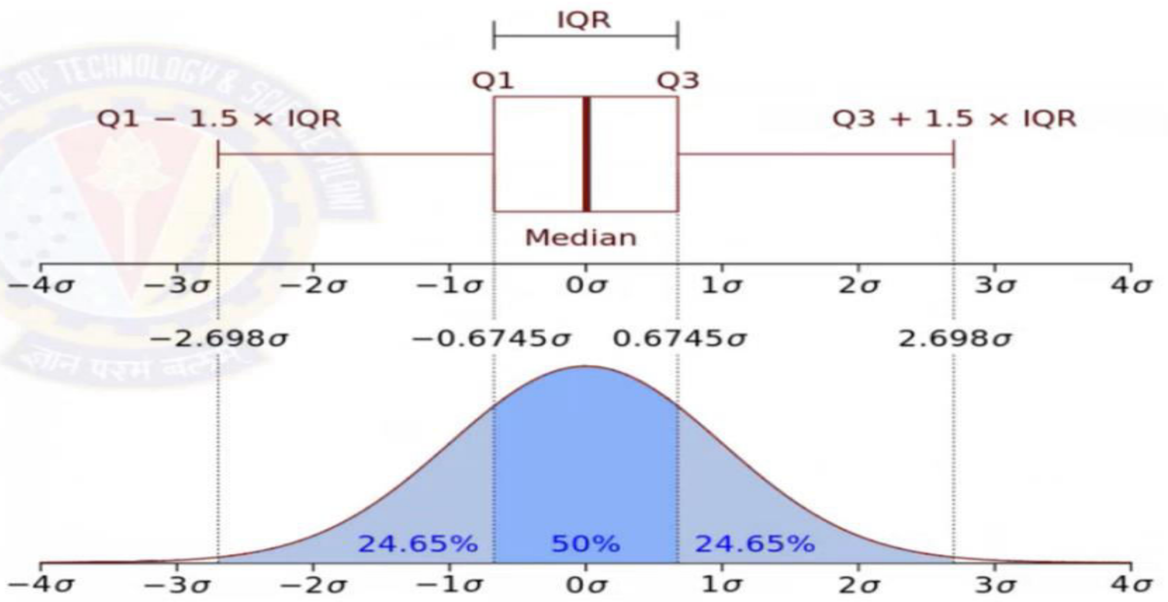
Lower limit = Q_1 - 1.5 x IQR

Upper limit = Q_3 + 1.5 x IQR

❖ Data points that lie below the lower limit or above the upper limit are potential outliers.

Outlier Detection





Statistical Summary

Cost	Weight	Weight1	Length	Height	Width	Width 1
count	159.000000	159.000000	159.000000	159.000000	159.000000	159.000000
mean	398.326415	26.247170	28.415723	31.227044	8.970994	4.417486
std	357.978317	9.996441	10.716328	11.610246	4.286208	1.685804
min	0.000000	7.500000	8.400000	8.800000	1.728400	1.047600
25%	120.000000	19.050000	21.000000	23.150000	5.944800	3.385650
50%	273.000000	25.200000	27.300000	29.400000	7.786000	4.248500
75%	650.000000	32.700000	35.500000	39.650000	12.365900	5.584500
max	1650.000000	59.000000	63.400000	68.000000	18.957000	8.142000

Axioms of Probability, Probability basics, Mutually exclusive and Independent events

Random Experiment

- Flip a coin. Did it land with heads or tails facing up?
- Walk to a bus stop. How long do you wait for the arrival of a bus?
- Give a lecture. How many students are listening?
- Transmit one of a collection of waveforms over a channel. What waveform arrives at the receiver?
- Transmit one of a collection of waveforms over a channel. Which waveform does the receiver identify as the transmitted waveform?

Random Experiment

- The term "**random experiment**" is used to describe any action whose outcome is not known in advance.

Examples:

- Counting how many times a certain word or a combination of words appears in the text of the "King Lear" or in a text of Confucius.
- Pulling a card from the deck.
- Monitor three consecutive phone calls going through a telephone switching office. Classify each one as a voice call (v) if someone is speaking, or a data call (d) if the call is carrying a modem or fax signal.

Sample Space

The sample space of a random experiment is a set S that includes all possible outcomes of the experiment.

Example:

- If the experiment is to throw a die and record the outcome, then, sample space is

  S = { 1,2,3,4,5,6}

- Manufacture an integrated circuit and test it to determine whether it meets quality objectives.

  The possible outcomes are “accepted” (a) and “rejected” (r).

  The sample space is S ={a , r}.

Event

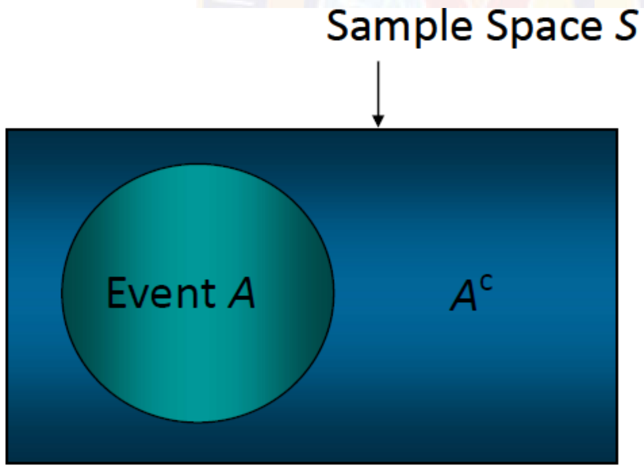
- **An event is a subset of the sample space of a random experiment.**

- An event is a set of outcomes of the experiment. This includes the *null* (empty) set of outcomes and the set of *all* outcomes.

- Each time the experiment is run, a given event *A* either *occurs*, if the outcome of the experiment is an element of *A*, or *does not occur*, if the outcome of the experiment is not an element of *A*.

Complement of an Event

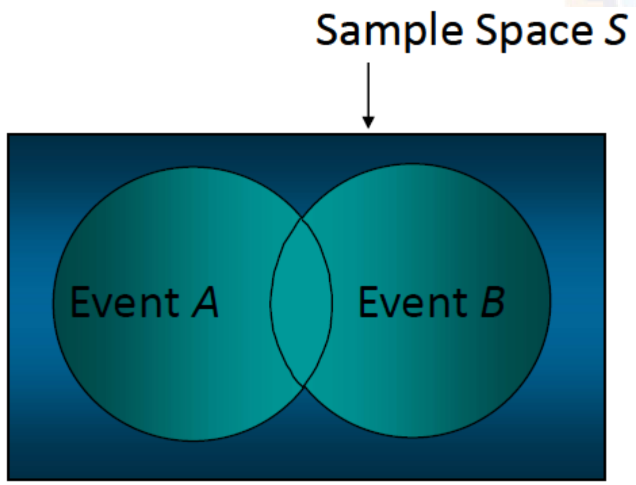
- The complement of event *A* is defined to be the event consisting of all sample points that are not in *A.*
- The complement of *A* is denoted by A^c.
- The Venn diagram below illustrates the concept of a complement.



Union & Intersection

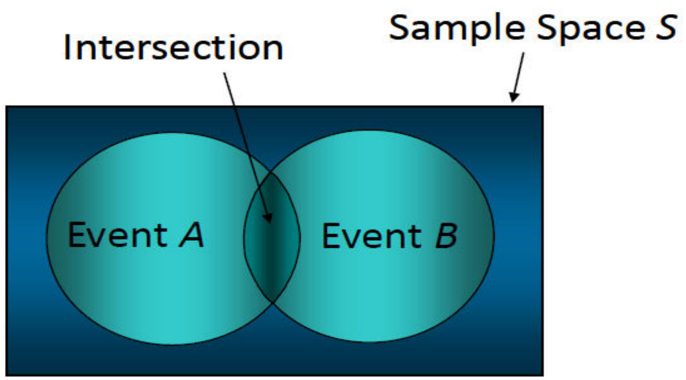
UNION

- The <u>union</u> of events *A* and *B* is the event containing all sample points that are in *A* or *B* or both.
- The union is denoted by *A* U *B*
- The union of *A* and *B* is illustrated below.



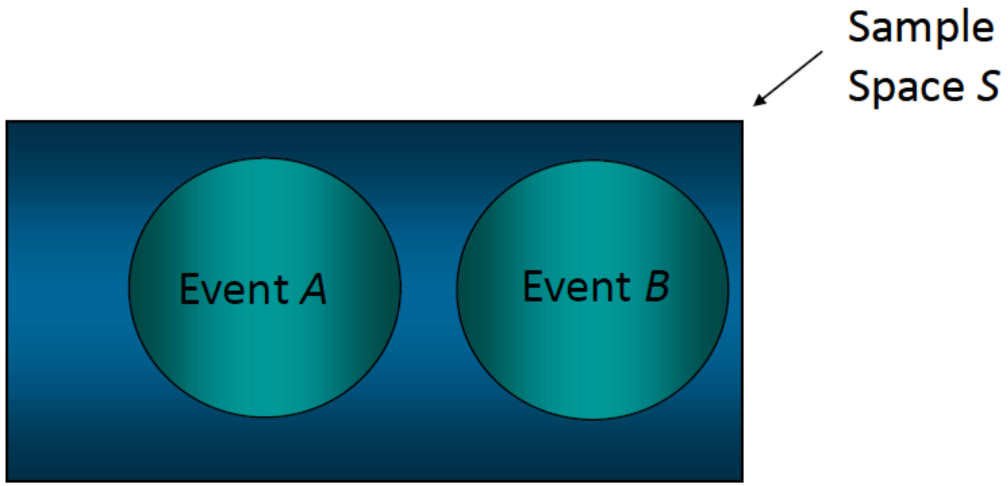
INTERSECTION

- The <u>intersection</u> of events *A* and *B* is the set of all sample points that are in both *A* and *B*.
- The intersection of *A* and *B* is the area of overlap in the illustration below.



Mutually Exclusive Events

- Two events are said to be mutually exclusive if the events have no sample points in common. That is, two events are mutually exclusive if, when one event occurs, the other cannot occur.
- **Example** : A simple example is flipping a coin and getting either heads or tails – you can't get both simultaneously.



Definition of Probability

Classical approach:

CLASSICAL PROBABILITY

$$\text{Probability of an event} = \frac{\text{Number of favorable outcomes}}{\text{Total number of possible outcomes}}$$

Empirical approach

Empirical or **relative frequency** is the second type of objective probability. It is based on the number of times an event occurs as a proportion of a known number of trials.

EMPIRICAL PROBABILITY The probability of an event happening is the fraction of the time similar events happened in the past.

In terms of a formula:

$$\text{Empirical probability} = \frac{\text{Number of times the event occurs}}{\text{Total number of observations}}$$

The empirical approach to probability is based on what is called the law of large numbers. The key to establishing probabilities empirically is that more observations will provide a more accurate estimate of the probability.

LAW OF LARGE NUMBERS Over a large number of trials, the empirical probability of an event will approach its true probability.

Axiomatic approach

Probability is a number that is assigned to each member of a collection of events from a random experiment that satisfies the following properties:

If S is the sample space and E is any event in a random experiment,

(1) $P(S) = 1$

(2) $0 \leq P(E) \leq 1$

(3) For two events E_1 and E_2 with $E_1 \cap E_2 = \varnothing$

$$P(E_1 \cup E_2) = P(E_1) + P(E_2)$$

Probability as a Numerical Measure of the Likelihood of Occurrence

Increasing Likelihood of Occurrence

Probability: 0 .5 1

The occurrence of the event is just as likely as it is unlikely.

The Addition Rule

- The probability that event *A* or *B* will occur is given by

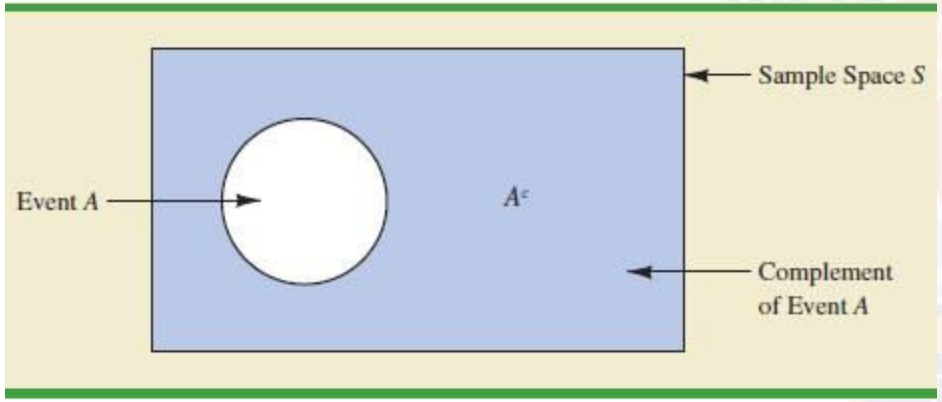
$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

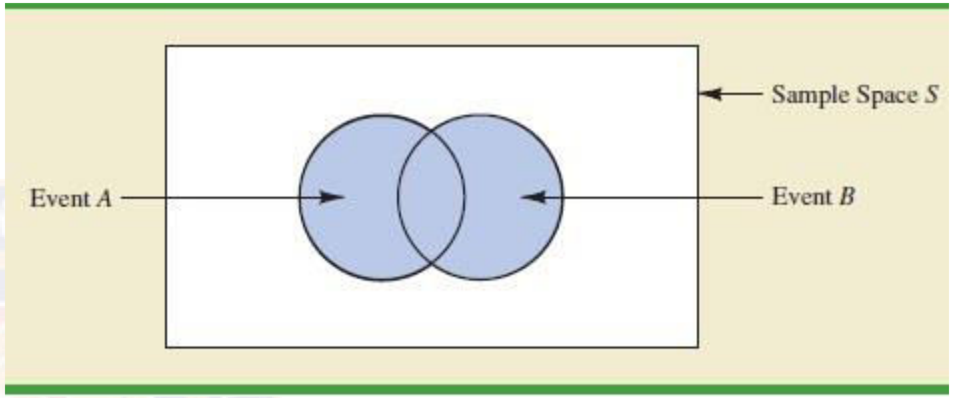
- If events *A* and *B* are **mutually exclusive**, then the rule is simplified as

$$P(A \cup B) = P(A) + P(B). \quad \{\text{Since, } P(A \cap B) = 0\}$$

Probability and Venn Diagram

$$P(A) = 1 - P(A^c)$$





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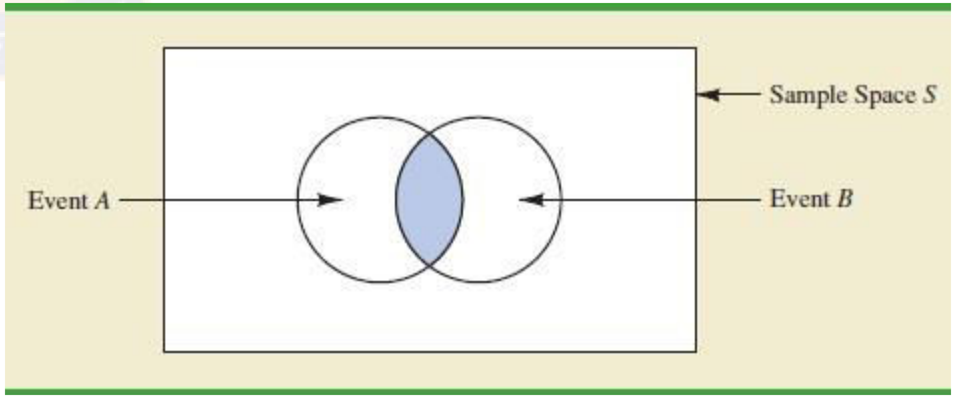


Independent & Dependent

Events are either

- *Independent* (the occurrence of one event has no effect on the occurrence of the other) or
- *Dependent* (the occurrence of one event gives information about the occurrence of the other)

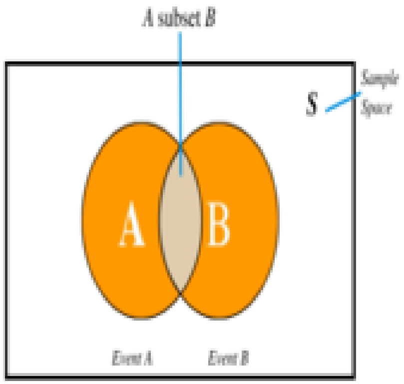
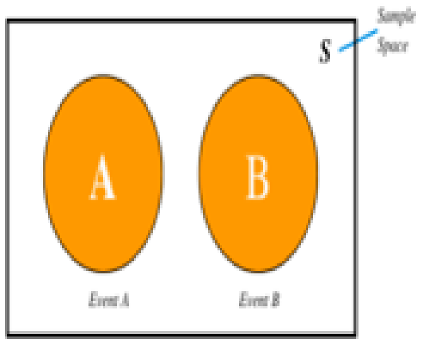
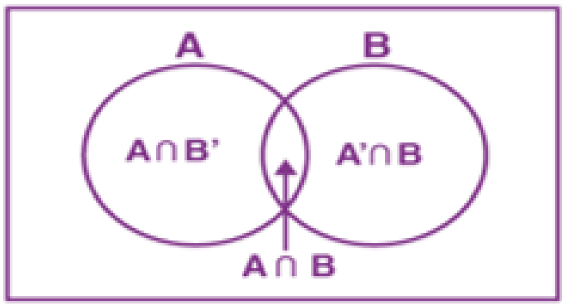
Example : Flipping a coin multiple times, where each flip is independent of the others.

- Independent events do not need to be mutually exclusive. Two events can be independent and still occur simultaneously.

A and B are independent if and only if (iff)

$$P(A \cap B) = P(A) \cdot P(B)$$

Comparison between Mutually Exclusive events and Independent Events

Comparison on the Basis	Mutually Exclusive Event	Independent Event
1. Definition	Here the events cannot happen simultaneously.	The occurrence and outcome of one event don't affect the occurrence and outcome of the other event.
2.Dependency	Occurrence of event A results in non-occurrence of event B.	Occurrence of event A does not affect event B in any manner.
3. Occurrence of both events	The mathematical formula for the representation of the mutually exclusive event is P(A⋂B) = 0	The mathematical formula for the representation of the mutually exclusive event is P(A⋂B) = P(A).P(B)
4. Venn Diagram Representation	  Here the events A and B do not overlap.	 Here the events do overlap.

Example 1

An experiment has the four possible mutually exclusive outcomes A, B, C, D. Check whether the following assignments of probability are permissible:

(a) $P(A) = 0.38$, $P(B) = 0.16$, $P(C) = 0.11$, $P(D) = 0.35$
(b) $P(A) = 0.31$, $P(B) = 0.27$, $P(C) = 0.28$, $P(D) = 0.16$
(c) $P(A) = 0.32$, $P(B) = 0.27$, $P(C) = -0.06$, $P(D) = 0.47$
(d) $P(A) = 1/2$, $P(B) = 1/4$, $P(C) = 1/8$, $P(D) = 1/16$
(e) $P(A) = 5/8$, $P(B) = 1/6$, $P(C) = 1/3$, $P(D) = 2/9$

Example 1

An experiment has the four possible mutually exclusive outcomes A, B, C, D. Check whether the following assignments of probability are permissible:

(a) P(A)= 0.38, P(B) = 0.16, P(C) =0.11, P(D) = 0.35 Permissible
(b) P(A)= 0.31, P(B) = 0.27, P(C) =0.28, P(D) = 0.16 NOT
(c) P(A)= 0.32, P(B) = 0.27, P(C) = -0.06, P(D) = 0.47 NOT
(d) P(A)= 1/2, P(B) = 1/4, P(C) = 1/8, P(D) = 1/16 NOT
(e) P(A)= 5/8, P(B) = 1/6, P(C) = 1/3, P(D) = 2/9 NOT

Example 2

- If two dice are thrown , what is the probability that the sum is

a) Greater than 8

b) Less than 6

c) Neither 7 nor 11

Solution

Probability of outcome, when a pair of die is thrown:

$$P(X = 2) = P\{(1, 1)\} = \frac{1}{36}$$

$$P(X = 3) = P\{(1, 2), (2, 1)\} = \frac{2}{36}$$

$$P(X = 4) = P\{(1, 3), (2, 2), (3, 1)\} = \frac{3}{36}$$

$$P(X = 5) = P\{(1, 4), (2, 3), (3, 2), (4, 1)\} = \frac{4}{36}$$

$$P(X = 6) = P\{(1, 5), (2, 4), (3, 3), (4, 2), (5, 1)\} = \frac{5}{36}$$

$$P(X = 7) = P\{(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)\} = \frac{6}{36}$$

$$P(X = 8) = P\{(2, 6), (3, 5), (4, 4), (5, 3), (6, 2)\} = \frac{5}{36}$$

$$P(X = 9) = P\{(3, 6), (4, 5), (5, 4), (6, 3)\} = \frac{4}{36}$$

$$P(X = 10) = P\{(4, 6) (5, 5), (6, 4)\} = \frac{3}{36}$$

$$P(X = 11) = P\{(5, 6), (6, 5)\} = \frac{2}{36}$$

$$P(X = 12) = P\{(6, 6)\} = \frac{1}{36}$$

Continued..

If two dice are thrown , what is the probability that the sum is

a) Greater than 8

a) $P(\text{sum} > 8)$

$$= P(9) + P(10) + P(11) + P(12)$$

$$= \frac{4}{36} + \frac{3}{36} + \frac{2}{36} + \frac{1}{36} = \frac{10}{36}$$

b)Less than 6:

b) $P(\text{sum} < 6)$

$$= P(5) + P(4) + P(3) + P(2)$$

$$= \frac{1}{36} + \frac{2}{36} + \frac{3}{36} + \frac{4}{36} = \frac{10}{36}$$

Continued

c) Neither 7 nor 11

$$P(\text{neither } 7 \text{ nor } 11)$$

$$= 1 - [P(7) + P(11)]$$

$$= 1 - \left[\frac{6}{36} + \frac{2}{36}\right]$$

$$= 28/36 = 7/9$$

Example 3

The probability that a student passes in statistics examination is 2/3 and the probability that he /she will not pass in mathematics examination is 5/9. The probability that he/she will pass in at least one of the examination is 4/5.

Find the probability that he /she will pass in both the examinations

Solution

The probability of passing in statistics P(S)=2/3

The probability of passing in Mathematics P(M)=1-5/9 =4/9

Probability of Passing at least one of these examination = P(S∪M) = 4/5

$$\boldsymbol{P(S \cup M) = P(S) + P(M) - P(S \cap M).}$$

$$\frac{4}{5} = \frac{2}{3} + \frac{4}{9} - P(S \cap M).$$

$$P(S \cap M) = \frac{2}{3} + \frac{4}{9} - \frac{4}{5} = \frac{14}{45}.$$

The probability of passing both examinations $P(S \cap M) = \frac{14}{45}$

Example 4

Suppose that 75% of all investors invest in traditional annuities and 45% of them invest in the stock market. If 85% invest in the stock market and/or traditional annuities, what percentage invest in both?

Solution

Let A be the event that a randomly selected investor invests in traditional annuities.

Let B be the event that he or she invests in the stock market.

Then $P(A) = 0.75$, $P(B) = 0.45$, and $P(A \cup B) = 0.85$

Since,

$$P(AB) = P(A) + P(B) - P(A \cup B)$$
$$= 0.75 + 0.45 - 0.85$$
$$= 0.35.$$

Example 5

Suppose the manufacturer’s specifications for the length of a certain type of computer cable are 2000 ± 10 millimeters. In this industry, it is known that small cable is just as likely to be defective (not meeting specifications) as large cable. That is, the probability of randomly producing a cable with length exceeding 2010 millimeters is equal to the probability of producing a cable with length smaller than 1990 millimeters. The probability that the production procedure meets specifications is known to be 0.99.

(a) What is the probability that a cable selected randomly is too large?

(b) What is the probability that a randomly selected cable is larger than 1990 millimeters?

Solution

Solution: Let M be the event that a cable meets specifications. Let S and L be the events that the cable is too small and too large, respectively. Then

(a) $P(M) = 0.99$ and $P(S) = P(L) = (1 - 0.99)/2 = 0.005$.

(b) Denoting by X the length of a randomly selected cable, we have

$$P(1990 \leq X \leq 2010) = P(M) = 0.99.$$

Since $P(X \geq 2010) = P(L) = 0.005$,

$$P(X \geq 1990) = P(M) + P(L) = 0.995.$$

This also can be solved

$$P(X \geq 1990) + P(X < 1990) = 1.$$

Thus, $P(X \geq 1990) = 1 - P(S) = 1 - 0.005 = 0.995$. $\lrcorner$

Example 6

Let S be a sample space and A and B are two mutually exclusive events such that AUB = S. If P(.) denotes the probability of the event, then what is the maximum value of P(A).P(B)?

Solution:

P(A) + P(B) = 1,since A and B are mutually exclusive {P(AUB) = P(S)=1}

When sum is a constant , product of two numbers becomes maximum when they are equal

So, P(A) = P(B) = ½

Hence P(A).P(B) = 1/4

Example 7

Let A & B denote the events with observations, when a pair of die are thrown, as follows. Are the events A and B independent?

Solution:

$A = \{(1, 2), (2, 1), (2, 2)\}$,

$B = \{(2, 2), (2, 3), (2, 4), (2, 5), (2, 6), (3, 2), (4, 2), (5, 2),(6, 2)\}$.

Therefore, the probabilities are

$P[A] = \frac{3}{36}$, $P[B] = \frac{9}{36}$, and $P[A \cap B] = P[(2, 2)] = \frac{1}{36}$

Clearly, $P[A \cap B] \neq P[A]P[B]$ and so A and B are dependent.

Example 8

A committee of 5 is chosen from a group of 8 men and 4 women. What is the probability that the group contains a majority of women?

Solution:

There are totally 12 people. Among them we have to choose 5 such that majority would be women. Hence the possible selection of men and women will be either:

1 men and 4 women or 2 men and 3 women.

$$P(1M \text{ and } 4\,W) = \frac{C_1^8 \times C_4^4}{C_5^{12}} = \frac{8}{792}$$

$$P(2M \text{ and } 3\,W) = \frac{C_2^8 \times C_3^4}{C_5^{12}} = \frac{112}{792}$$

Therefore the probability of selecting 5 people is $\frac{8}{792} + \frac{112}{792} = \frac{5}{33}$

Practice problems

Q1: A Survey conducted by a bank revealed that 40% of the accounts are savings accounts and 35% of the accounts are current accounts and the balance are loan accounts.

- What is the probability that an account taken at random is a loan account ? Ans: 0.25
- What is the probability that an account taken at random is **NOT** savings account ? Ans: 0.60
- What is the probability that an account taken at random is **NOT** a current account ? Ans: 0.65
- What is the probability that an account taken at random is a current account or a loan account? Ans: 0.60

Practice Problems

Q2. A speaks truth in 80% cases and B speaks in 60% cases. What percentage of cases are they likely to contradict each other in stating the same fact.

- Hint / Ans: *P*(Contradiction)=P(A truth and B false)+P(A false and B truth)=0.32+0.12=0.44

Q3. In a certain residential suburb, 60% of all households get Internet service from the local cable company, 80% get television service from that company, and 50% get both services from that company. If a household is randomly selected,

i. What is the probability that it gets at least one of these two services from the company?

Ans: P(Internet or Television)=0.90

ii. What is the probability that it gets exactly one of these services from the company?

Ans: *P*(Exactly one service)=*P*(Internet but not Television)+*P*(Television but not Internet)=0.10+0.30=0.40

Practice problems

Q4: The next generation of miniaturised wireless capsules with active locomotion will require two miniature electric motors to manoeuvre each capsule. Suppose 10 motors have been fabricated but that, in spite of test performed on the individual motors 2 will not operate satisfactorily when placed into capsule, to fabricate a new capsule,2 motors will be randomly selected(that is, each pair of motors has the same chance of being selected) find the probability that

a) Both motors will operate satisfactorily in the capsule.

b) One motor will operate satisfactorily and other will not.

Practice problems

Q5. Suppose that 55% of all adults regularly consume coffee, 45% regularly consume carbonated soda, and 70% regularly consume at least one of these two products.

i) What is the probability that a randomly selected adult regularly consumes both coffee and soda?

ii) What is the probability that a randomly selected adult doesn't regularly consume at least one of these two products?

Practice problems

Q6. Suppose a student is selected at random from 80 students where 30 are taking mathematics, 20 are taking chemistry and 10 are taking both. Find the probability 'p' that the student is taking Mathematics or chemistry?.

Q7. If A and B are events with P(A U B) = 7/8,

P(A ∩ B) = ¼ and P(A') = 5/8, find P(A), P(B) and P(A ∩ B')**).**

Q8. In a Sample space, events A and B are such that P(A)=P(B), $P(\bar{A} \cap \bar{B}) = P(A \cap B) = \frac{1}{6}. Find$

$a)\ P(A),$

$b) P(\bar{A} \cup \bar{B})$

$c)\ P(\text{exactly one of the events } A \text{ or } B)$

Practice problems

Q9. An insurance company offers four different deductible levels—none, low, medium, and high—for its homeowner's policyholders and three different levels—low, medium, and high—for its automobile policyholders. The accompanying table gives proportions for the various categories of policyholders who have both types of insurance.

For example, the proportion of individuals with both low homeowner's deductible and low auto deductible is .06 (6% of all such individuals).

Auto	N	L	M	H
L	0.04	0.06	0.05	0.03
M	0.07	0.10	0.20	0.10
H	0.02	0.03	0.15	0.1

Suppose an individual having both types of policies is randomly selected.

a. What is the probability that the individual has a medium auto deductible and a high homeowner's deductible?

b. What is the probability that the individual has a low auto deductible? A low homeowner's deductible?

c. What is the probability that the individual is in the same category for both auto and homeowner's deductibles?

d. Based on your answer in part (c), what is the probability that the two categories are different?

e. What is the probability that the individual has at least one low-deductible level ?

f. Using the answer in part (e), what is the probability that neither deductible level is low ?

Summary

Thank You!